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	RV College of Engineering® **Department of Computer Science and Engineering** **CIE - II**

Course & Code	**PARALLEL ARCHITECTURE AND GPU PROGRAMMING (CS372IA)**			**Semester: VII**
Date: 15/12/2025	**Duration:** 120 minutes	**Test Max. Marks**: 50 Marks **Quiz Max Marks:** 10 Marks		**Faculty In Charge:** AN/SANS/MMN
USN :	**Name :**			

NOTE: ***Answer all the questions***

Sl.no.	Questions	Marks	* BT	*CO
	QUIZ- 2			
1	The VMIPS instruction for gather and scatter are ______ and ________	2	2	1
2	What is vector register chaining?	2	1	1
3	Define loop-carried dependence, and why does it reduce loop-level parallelism?	2	1	1
4	Why is sequential consistency simple to understand but difficult to implement efficiently?	2	2	2
5	Define a warp	2	1	1
	TEST-2			
1.a	Examine the following synchronization primitives with the pseudo code: i. LoadLinked and Store Conditional ii. Spinlocks	06	2	1
b.	Discuss various Relaxed Consistency models.	04	2	1
2.a	Write a CUDA program for Vector Addition	10	3	3
3.a	Illustrate with a state transition diagram for an individual cache block in a directory based protocol.	10	3	2
4.a	Write and explain the strip-mined version of the DAXPY loop in C.	04	4	3
b	Why most vector processors use memory banks. Cray T90 has 32 processors, each capable of generating 4 loads and 2 stores per clock cycle. The processor clock cycle is 2.167 ns, while the cycle time of the SRAMs used in the memory system is 15 ns. Calculate the minimum number of	06	3	4

	memory banks required to allow all processors to run at full memory bandwidth.			
5.a	Discuss the memory structures of an NVIDIA GPU.	04	3	2
b	Explain kernel invocation and execution model of CUDA	06	2	4

COURSE OUTCOMES:

CO1	Understand core concepts of computer architecture to investigate parallel algorithms for solving computational problems.
CO2	Analyze the performance of parallel programming models and techniques and develop proficiency in parallel algorithm design using MPI, OpenACC, and CUDA.
CO3	Design and implement parallel algorithms using appropriate parallel programming models
CO4	Demonstrate Parallel computing concepts for suitable compute intensive real time applications

	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4
Marks	6	20	30	04	-	-	18	16	14	12

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